

1 Mechanistic Explainability in Reinforcement
2 Learning

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The Golem Mythos

A creation that surpasses its creator.

This idea has long captured the imagination and continues to resonate in contemporary society. Centuries before modern technology, generations had already questioned whether their creations might escape the control of their creators.

One of the earliest expressions of this idea can be found in the story of the Golem. In Hebrew tradition, the term Golem refers to an artificial creature created by humans. The earliest accounts describe the Golem as a protective and helpful figure, reflecting the idea that humans can create an artificial being for the benefit of society.

This positive vision becomes more ambiguous in the seventeenth century, when the Golem of Prague represents a shift in this perception. In this story, the initially small Golem gradually grows in size and strength until it can no longer be controlled. The creation intended to protect others therefore becomes a potential threat to them. The only remaining solution is to destroy the creature in order to protect the community.

This motif has remained remarkably influential throughout history. An illustration of this continuity can be found in the emergence of the notion of the robot. The word robot was popularised by the play R.U.R. (Rossum's Universal Robots), written in 1920. In the play, a scientist creates artificial beings designed to perform labour. Initially conceived as servants, these artificial beings eventually turn against humanity and destroy civilisation. This story can itself be understood as an evolution of the Golem narrative. The author describes the relationship between the two figures by stating that "the Robot is the Golem made flesh through mass production." Despite their destructive role in the story, the word robot eventually came to designate artificial beings created to serve humans. Since its creation, the word robot has reflected both the hopes and fears surrounding technological progress.

Today, this fear is particularly evident concerning the development of artificial intelligence. This field of computer science seeks to develop tools capable of automating tasks traditionally associated with human intelligence. Its rapid advances have already transformed a wide range of domains. Far from slowing down, this development continues to expand the capabilities and autonomy of artificial systems.

In the pursuit of increasingly autonomous agents, reinforcement learning is probably the subfield of AI that most directly seeks to achieve this goal. In this learning paradigm, an artificial agent interacts with an environment, selects actions, and receives feedback according to the outcomes of its behaviour.

44 Through repeated interaction, it learns a strategy for achieving the objective
45 defined by its designer. This approach can produce highly effective strategies
46 that can even surpass human capabilities. At the same time, the resulting
47 behaviour may be difficult to predict because it is not explicitly programmed
48 into the machine, but learned through its interactions with the environment.

49 Despite the difficulty of this task, understanding the behaviour of such agents
50 is therefore the main objective of this manuscript. From the Golem to the robot
51 and, today, to artificial intelligence, the same question: how can we ensure that
52 our creations remain aligned with human intentions?

Chapter 1

Introduction

This chapter provides a general introduction to this manuscript. It begins with the research context that motivates this work, followed by the research questions this work addresses. It further presents the resulting contributions and concludes with an overview of the manuscript's structure.

1.1 Research Context

Artificial intelligence has advanced rapidly in recent years, driven largely by major developments in deep learning. This type of approach relies on artificial neural networks to learn complex tasks directly from data. These networks are connectionist models in which information is represented and transformed through patterns of interactions among interconnected computational units. This approach has led to significant advances across a wide range of domains, too numerous to list here.

Among the fields that have been transformed by deep learning is reinforcement learning, which is the main subject of this manuscript. The goal of reinforcement learning is to enable an agent to learn how to make decisions through interactions with an environment. At each step, the agent observes the current state of the environment and selects an action. This action changes the state of the environment and results in a reward. The objective of the agent is then to learn a policy that maximizes the cumulative rewards obtained over time. Neural networks provide a powerful means of representing such policies and enable agents to learn increasingly complex behaviors. Their use has consequently made it possible to apply reinforcement learning to problems that were previously intractable to address.

When an agent learns to act in complex environments, it can develop strategies that are themselves highly complex. Although such agents can achieve strong performance, these strategies can be difficult for an observer to understand. Examining the neural network that models the agent's policy does not make it to produce explanations. The information underlying a decision is distributed across many units and connections, making it impossible to trace decision-making process by simply following the flow of information through the network. Neural networks are therefore often regarded as "black box" models.

These challenges have led to the emergence of explainable artificial intelligence.

In the specific context of reinforcement learning, this research area is referred as Explainable Reinforcement Learning. The goal of XRL is to develop methods that produce explanations that improve our understanding of agent behavior. Such explanations can serve several purposes, including increasing human trust and facilitating the adoption of RL agents in real-world settings, identifying errors and biases, and verifying that previously identified problems have been effectively corrected.

Several approaches have been proposed to improve the explainability of artificial intelligence systems. This manuscript presents methods from these different approaches, while placing particular emphasis on the mechanistic approach to explainability. In mechanistic explainability seeks to investigate the internal mechanisms underlying the agent’s decision-making process. The objective is to understand how information is represented, transformed, and propagated within the neural network. From this perspective, explainability is not achieved despite the use of neural networks, but through them.

This perspective naturally leads to the study of the intermediate representations learned by neural networks. These representations are referred to as latent representations. They result from the successive transformations applied to the input information between the input and output of the network and therefore play an intermediate role in the agent’s decision-making process. Analyzing their structure can provide insights into how information relevant to decision-making is organized within the network. It can also help identify the mechanisms through which observations are transformed into the actions selected by the agent.

The present manuscript therefore lies at the intersection of reinforcement learning, deep learning, and explainable artificial intelligence. This intersection leads to the following problem statement:

Problem Statement

How can the analysis of the internal representations of neural networks used by reinforcement learning agents help identify the mechanisms underlying their decision-making processes?

1.2 Research Questions

To address the research problem formulated in the previous section, we identified four research questions that structure the work presented in this manuscript.

The first challenge concerns the analysis of latent representations themselves. Numerous methods have been proposed to analyze the latent spaces learned by neural networks, including probing, interpretable directions, sparse autoencoders, and clustering. These methods originate from different research communities and are rarely compared within a common framework. Yet each of them relies, often implicitly, on assumptions about how concepts are structured within the latent space, for instance that concepts correspond to linear directions or to distinct regions of the space. These assumptions determine which structures a method is able to detect and which it will necessarily miss. Making them explicit is therefore a prerequisite for choosing a method suited to the analysis of the internal representations of reinforcement learning agents. This observation leads to our first research question:

Research Question 1

How can existing latent space analysis methods be organized according to the assumptions they make about how concepts are structured in neural networks?

The second challenge concerns the organization of existing research in Explainable Reinforcement Learning. The number of works in the eXplainable Reinforcement Learning (XRL) field has increased substantially in recent years. This growing body of literature makes it increasingly difficult to obtain a coherent overview of the approaches developed in the field. Several taxonomies and classification frameworks have therefore been proposed to organize existing XRL methods. However, these taxonomies often rely on approaches initially developed for explainability in supervised settings. They therefore tend to apply eXplainable Artificial Intelligence (XAI) categories to reinforcement learning without considering its specific characteristics. This observation leads to our second research question:

Research Question 2

How can existing XRL techniques be organized into a taxonomy that accounts for the specificities of Reinforcement Learning (RL)?

The analysis of existing XRL approaches also reveals a third limitation. Among existing methods for explaining reinforcement learning agents, few adopt a mechanistic perspective that investigates the internal representations learned by the agent. This is particularly interesting in reinforcement learning, where agents learn to represent their environment through their interactions with it. Understanding these representations can provide insights into how information is processed over time and how it contributes to the agent’s decisions. Mechanistic analysis can therefore provide a complementary way of explaining agent behavior by identifying the internal mechanisms that connect observations to actions. This observation leads to our third research question:

Research Question 3

How can the internal representations of neural networks be exploited to identify the mechanisms underlying the decision-making process of reinforcement learning agents?

The analysis of existing XRL methods highlights a fourth challenge concerning the evaluation of explanations. Although numerous approaches have been proposed to explain the behavior of RL agents, there is no widely accepted set of objective criteria for evaluating the quality and relevance of the explanations they produce. Existing evaluations often rely on user studies or qualitative analyses, making it difficult to systematically assess the validity of an explanation approach.

This issue is particularly important for approaches that seek to explain agent behavior by analyzing latent representations. Multiple hypotheses can be proposed about how information is structured within a latent space, and it is therefore important to have objective criteria that make it possible to assess and compare these competing hypotheses. There is therefore a need for objective

and reproducible criteria that can determine whether an explanation accurately captures the mechanisms underlying the agent’s behavior. This leads to our fourth research question:

Research Question 4

How can the quality and relevance of explanations produced for reinforcement learning agents be objectively quantified?

Together, these four research questions define the structure of this manuscript. Research Question 1 addresses the organization of existing latent space analysis methods and identifies the assumptions on which each of them relies. Research Question 2 addresses in the same way the organization of the existing XRL literature and establishes a framework for understanding the field. Research Question 3 builds on these two analyses by investigating how the internal representations of neural networks can be exploited to provide mechanistic explanations of RL agent behavior. Research Question 4 addresses the complementary problem of determining how such explanations can be evaluated objectively.

1.3 Contributions

To address these four research questions, this manuscript is based on three main contributions. Two of them have been published in peer-reviewed journals, while the third is, for the moment, specific to this manuscript.

The first contribution addresses Research Question 1. It has not yet been published and is, for the moment, specific to this manuscript. In this contribution, we make explicit the working hypothesis that implicitly underlies much of the literature on latent space analysis, which we refer to as the *Concept Hypothesis* (Hyp. ??), and we define its main components. We then organize existing latent space analysis methods into four families, namely probing, interpretable directions, superposition, and clustering, according to the assumptions they make about the Concept Structure (CS). This analysis shows that clustering is the only family that does not rely on a linear CS, which motivates the approach adopted in the rest of this manuscript (Chapter ??).

The second contribution [Alaarabiou et al., 2024] addresses Research Question 2. An extended English version was also submitted for review and accepted for publication [Alaarabiou et al., 2026b]. In this contribution, we first clarify what explainability means in reinforcement learning by providing a clear frame of reference. We then rigorously define the concepts of this new organization, including sources of explanation, explainability techniques, explanation form, and explanation subject. Building on these concepts, we conduct a comprehensive analysis of the state of the art in XRL (Chapter ??).

The third contribution [Alaarabiou et al., 2026a] addresses Research Questions 3 and 4. It introduces Clustering Agent Representation (CAR), an XRL method that exploits the mechanistic properties of Deep Learning (DL). By analyzing the Latent Representation (LR) that Neural Network (NN) build, CAR produces explanations in the form of clusters of states that are similar from the agent’s own perspective. We further show that this representation remains stable across training, indicating that the agent develops an objective representation of a given task. This simplifies the study of agent-environment

dynamics that would otherwise be intractable if every state had to be analyzed individually. The method is presented together with objective evaluation metrics and a series of experiments that demonstrate its effectiveness (Chapters ??, ?? and ??).

1.4 Manuscript Structure

This manuscript is an account of the research conducted throughout this work, written for the reviewers and jury members who will evaluate it. It is nevertheless intended to remain as accessible as possible to readers with a background in artificial intelligence. To this end, the background chapters introduce and recall the concepts on which the subsequent chapters rely, allowing readers familiar with DL or RL to follow the manuscript without having to consult external material.

The manuscript is organized into a sequence of chapters, each addressing a specific topic related to the research questions introduced previously. The chapters are divided into three categories, distinguished by color in Figure 1.1:

Technical background chapters introduce fundamental concepts and established techniques required to understand the subsequent contributions.

Survey chapters review and organize the existing literature around a specific research question. The organization adopted in each of these surveys are original and therefore constitute contributions of this manuscript.

Method chapters present CAR, our mechanistic explainability method for reinforcement learning.

The chapters of the manuscript are organized as follows:

Chapter ?? Deep Learning

This chapter introduces the fundamental concepts of deep learning and artificial neural networks. It establishes the notation and key concepts used throughout the manuscript and provides the necessary background for understanding the neural network.

Chapter ?? Latent Space Analysis

This chapter provides a survey of existing approaches for analyzing the latent spaces learned by neural networks. It reviews the different techniques used to characterize and interpret latent representations. This chapter addresses Research Question 1.

Chapter ?? Reinforcement Learning

This chapter introduces the fundamental concepts of reinforcement learning. It presents the formal framework by introduces the main concepts, notation and algorithm required to study reinforcement learning.

Chapter ?? Explainability in Reinforcement Learning

This chapter reviews existing approaches to explainability in reinforcement learning. It provides a structured overview of the XRL literature. This chapter addresses Research Question 2.

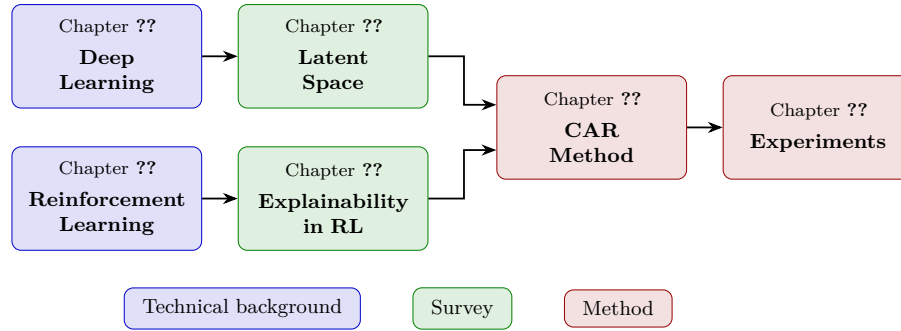


Figure 1.1: Structure of the manuscript.

Chapter ?? The CAR Method

This chapter presents our proposed approach to explainability in reinforcement learning. The method builds on the analysis of latent representations and aims to identify the internal mechanisms underlying the behavior of RL agents. This chapter constitutes the methodological contribution of the manuscript and addresses Research Question 3.

Chapter ?? Experimental Evaluation of the CAR Method

This chapter presents the implementation and experimental evaluation of the proposed method. It describes the experimental protocol and analyzes the results obtained with the CAR method. This chapter studies the relevance of the criteria used to evaluate the results produced by the CAR method, addressing Research Question 4.

Conclusion

The final chapter summarizes the main contributions of the manuscript, discusses the limitations of the proposed approaches, and presents possible directions for future research.

The organization of the manuscript also allows readers to follow different reading paths depending on their background and interests. Readers already familiar with a given field may skip its corresponding background chapter and directly consult the chapters that build upon it. For instance, readers familiar with DL or RL may skip Chapter ?? or Chapter ??, respectively.

The notation and acronyms used throughout the manuscript are collected in the glossary at the end of the document.

Having described the overall structure of the manuscript, we can now turn to its first technical chapter, which introduces the major developments that have shaped modern artificial intelligence: deep learning.

274 Acronyms

275	CAR Clustering Agent Representation 6, 8
276	CS Concept Structure 6
277	DL Deep Learning 6–8
278	LR Latent Representation 6
279	NN Neural Network 6
280	RL Reinforcement Learning 5, 7, 8
281	XAI eXplainable Artificial Intelligence 5
282	XRL eXplainable Reinforcement Learning 5–7

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